

Capstone Project: Optimizing Streaming Service Performance Through User Data Insights

Abstract

In the competitive streaming industry, platforms like Netflix thrive by understanding viewer behavior. This project analyzes user data to uncover meaningful patterns in content engagement, subscription trends, device usage, and churn risk. By leveraging Excel, SQL, and Power BI, we provide actionable insights to improve recommendation systems, enhance user retention, and optimize marketing and operational strategies.

Project Goal

To identify patterns and trends in user activity, watch behavior, and subscription metrics that can help Netflix:

- Enhance personalized content recommendations
- Improve user retention and reduce churn
- Strategically tailor content based on region, device, and subscription type
- Identify and reward high-value users

Tools & Technologies Used

- Microsoft Excel: Data cleaning, aggregation, pivot tables
- MySQL: Data querying and analysis

- Power BI: Dashboard creation and data visualization.

Dataset Overview

The dataset contains structured information about user interaction with the Netflix platform. Key features include:

- User ID, Subscription Type, Name, Age,
- Watch Time, Start month, Favourite Genre, Revenue per Hour
- Device Type, Country, Subscription duration
- Engagement metrics and churn status

Key Power BI Dashboard Insights

- Top 5 Countries by Watch Time: Countries with the highest cumulative watch time help target regional content development and marketing.
- User Engagement Analysis: Categorized users based on high/medium/low engagement using watch hours and session frequency.
- Top Genres by Watch Time: Identified the most-watched genres, guiding future content investments.
- Device Usage: Analyzed preferred streaming devices (TV, mobile, desktop) to enhance UX on those platforms.
- Favourite genre by Subscription Plan: Compared content types (Movies, Series, Documentaries) across subscription tiers (Basic, Standard, Premium).
- Average Subscription Duration per Type: Evaluated which subscription type has the longest average tenure to focus retention efforts.

- Users with Longest Subscriptions: Profiled long-term subscribers to identify satisfaction drivers.

- Device Usage by Watch Time: Correlated watch time with device type to refine device-specific engagement strategies.

- Most Popular Content Recommendation: Used aggregated watch counts and ratings to generate content recommendations.

SQL Queries Used

use netflix;

select *from netflix;

1. #top 5 country by watch time

```
SELECT  country,    SUM(watch_time_hours)  AS  
total_watch_time FROM netflix  
GROUP BY country
```

```
ORDER BY total_watch_time  
DESC LIMIT 5;
```

2. #user engagement analysis

```
SELECT  is_high_engagement, COUNT(*) AS  
user_count FROM netflix  
GROUP BY is_high_engagement;
```

3. # top genres by watch time

```
SELECT favorite_genre, SUM(watch_time_hours) AS  
total_hours FROM netflix  
GROUP BY  
favorite_genre ORDER  
BY total_hours DESC;
```

4. #device usage

```
SELECT device, COUNT(*) AS  
users FROM netflix  
GROUP BY device  
ORDER BY users  
DESC;
```

5. #content type preference by subscription plan

```
SELECT subscription_type, content_type_recommendation,  
COUNT(*) AS count FROM netflix  
GROUP BY subscription_type,  
content_type_recommendation ORDER BY  
subscription_type, count DESC;
```

6. #churn risk users (low engagement + not loyal)

```
SELECT user_id, name, watch_time_hours, is_loyal_customer,  
is_high_engagement FROM netflix  
WHERE is_loyal_customer = 'No' AND is_high_engagement = 'No';
```

7. #avg subscription duration per subscription type

```
SELECT      subscription_type,  round(AVG(subscribed_duration),2)
            AS avg_subscription_duration

FROM Netflix

GROUP BY subscription_type;
```

8. #user with long
subscription SELECT
user_id, name,
subscribed_duration FROM
netflix

```
ORDER BY
subscribed_duration DESC
LIMIT 10;
```

9. #device usage by
watch time SELECT
device,

```
ROUND(AVG(watch_time_hours), 2) AS
avg_watch_time FROM netflix
GROUP BY device
```

```
ORDER BY avg_watch_time DESC;
```

10. #most popular content recommendation

```
SELECT    content_type_recommendation,    COUNT(*)    AS  
recommendation_count FROM netflix  
GROUP  
content_type_recommendation  
ORDER BY recommendation_count  
DESC;
```

11. #engagement vs

```
loyalty SELECT  
is_high_engagement,  
is_loyal_customer,  
COUNT(*)    AS  
user_count  
FROM netflix  
  
GROUP BY is_high_engagement, is_loyal_customer;
```

Insights Derived

Content Recommendations

- Personalized recommendations should prioritize top genres like Action, Drama, and Comedy, featuring high-rated, high-completion content to enhance viewer engagement and satisfaction.

Subscription Retention

- To boost retention, focus on improving early user engagement through targeted onboarding content and regional content drops, especially for demographics and areas with higher churn.

Peak Streaming Hours

- Peak viewing occurs during evening hours and weekends, suggesting an ideal window for major content releases and promotional notifications to maximize reach and impact.

Marketing Strategies

- Marketing efforts should be content-centric and segmented by age group and platform, leveraging high-performing campaigns and referrals to optimize subscriber growth and ROI.

Yearly Revenue Performance

- The streaming service has achieved a **total yearly revenue of 375K**, highlighting significant business traction and user engagement.

User Distribution by Country

- **Germany, Australia, and the USA** are the **top 3 countries by watch hours**, showing these regions are highly active and engaged markets.

Device Usage Patterns

- Device usage is well-balanced across **Laptop, Tablet, Smartphone, and Smart TV**, each contributing roughly **25%**. This indicates a need for consistent cross-device performance optimization.

Monthly Revenue Trend

- Revenue peaks around **June and November**, suggesting effective marketing campaigns or content drops during these months, possibly tied to summer and holiday season viewing trends.

Genre and Content Preference

- **Drama, Horror, and Documentary** are the **top 3 genres** by user preference and watch time, with **Drama** being the most favored across **countries and subscription types**.

Subscription Type & Engagement

- The user base is fairly distributed across **Basic (33%), Standard (33%), and Premium (34%)** plans. Premium users show slightly higher **average subscription durations**, and there is a high count of **loyal and engaged customers**, indicating strong retention.

Conclusion

The analysis of Netflix & streaming service data reveals key insights into user behavior, content preferences, and revenue patterns. With a yearly revenue of 375K and a loyal user base, Drama, Horror, and Documentary emerge as top genres, while Germany and Australia lead in watch time. Device usage is evenly split across platforms, and monthly revenue peaks in mid-year months. Subscription types are well-distributed, with Premium users showing longer engagement. These findings can help tailor content, optimize user experience, and boost strategic decision-making for growth.

